INCH-POUND

MIL-S-22432/9E

28 November 1990

SUPERSEDING

MIL-S-22432/9D

17 January 1975

MILITARY SPECIFICATION SHEET

SERVOMOTOR, TYPE 18SM4C

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Servomotors described herein shall consist of this specification sheet and the issue of the following specification listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-22432.

TABLE I. Requirements.

Requirement	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	hz	
Test	400	hz	±1%
Range	380 to 420	hz	
Voltage:			
Reference phase, rated	115	volts rms	
Reference phase, test	115	volts rms	±1%
Reference phase, range	103.5 to 126.5	volts rms	
Control phase, rated (series)	115	volts rms	
Control phase, test (series)	115	volts rms	±1%
Control phase, range (series)	0 to 126.5	volts rms	
Control phase, rated (parallel)	57.5	volts rms	
Control phase, range (parallel)	0 to 63.25	volts rms	
Current:			
Reference phase	0.200	amperes rms	maximum
Control phase (series)	0.200	amperes rms	maximum
Control phase (parallel)	0.400	amperes rms	maximum
Power:			
Reference phase	9.1	watts	maximum
Control phase (series)	9.1	watts	maximum
Control phase (parallel)	9.1	watts	maximum
Synchronous Speed	6,000	rpm	
impedance:			
Reference phase	268+j580	ohms	±30%
Control phase (series)	268+j580	ohms	±30%
Power Factor:			
Reference phase	0.42	--	±1%
Control phase (series)	0.42	--	±1%
Electrical breakaway torque	3.0	volts rms	maximum
Coupling in control winding	0.7	ratio	minimum

AMSC - N/A

FSC 6105

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:16Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements - Continued.

Requirements	Value	Unit	Tolerance
Mechanical:			
No load speed	4,800	rpm	minimum
Stall torque at 23°	2.20	oz.-in.	minimum
Rotor moment of inertia	4.0	$gm-cm^2$	maximum
Radial Play	0.0008	inch	maximum
End Play	0.0045	inch	maximum
	0.0020	inch	minimum
Temperature Rise	85.0	degrees C	maximum
Time Constant	0.0200	seconds	maximum
Reversing Time	0.0300	seconds	maximum
Maximum Power Output	1.75	watts	minimum
Weight	13.5	ounces	maximum

NOTE:

1. Type 18SM4C servomotors shall be capable of operation in ambient temperatures ranging from -55°C to 100°C.

TABLE II. Part identifying number variant characteristics.

Part Identifying Number 1/	A ±.010	C +.003 -.005	Shaft Style	Terminal End	G Maximum
M22432/9-01C	436	1.750	Splined	Terminal board	1.875
M22432/9-02C	.540	1.750	Splined	Terminal board	1.875

1/Part identifying number M22432/9-01C shall be the NATO standard

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:16Z
Check the source to verify that this is the current version before use.

TABLE III. Part identifying number cross-references. 1/

SUPERSEDED PART NUMBERS		NEW PART IDENTIFYING NUMBER
MIL-S-22432/9D	BUORD Drawings	
M22432/9-001	675057 MK 8 MOD 1	M22432/9-01C
M22432/9-002	675029 MK 8 MOD 0	M22432/9-02C

1/ All line items shown in Table III refer to equivalent and interchangeable servomotors of the same type designation modification. Part number changes do not affect form, fit, or function of the servomotors listed therein.

Custodians:
Army - AR
Navy - AS
Air Force - 11

Review Activities:
Army - MI, ME, ER
Navy - SH, OS, EC
Air Force - 99
DLA - GS

User Activities:
Army - AV
Navy - MC, CG

Preparing Activity:
Navy - AS

(Project 6105-0162-03)

Source: https://assist.dla.mil -- Downloaded: 20261O-19T12:16Z
Check the source to verify that this is the current version before use.